

Joan Alcaide-Núñez

Email: joan.alcaide@campus.lmu.de · Website: joanalnu.github.io · GitHub: github.com/joanalnu

EDUCATION

Ludwig-Maximilians-Universität (LMU) Munich B.Sc. in Physics <i>Core concentration in foundational and mathematical physics pathways.</i>	Munich, Germany Oct 2025 – Present
German School of Barcelona Dual High School Diploma (Germany + Spain) <i>Academic Achievement Award with a perfect Grade Point Average of 1.0 / 1.0.</i>	Barcelona, Spain Sept 2017 – May 2025
Advanced Training: Gravity and Black Holes (<i>Perimeter Institute GoPhysics!</i> , 2025) · Mathematics, Statistics, and Data Science (<i>Autonomous University of Barcelona</i> , 2022) · Algorithms and C++ Programming (<i>UPC / Harbour Space</i> , 2022–2023).	

RESEARCH EXPERIENCE

Theoretical Probes of Spacetime Independent Research Project	Remote / Independent Study July – Sept 2026
- Derived Modified Dispersion Relations (MDR) for high-energy photons traveling through non-standard cosmological backgrounds to probe potential Planck-scale Lorentz invariance violations. - Applied Hamilton-Jacobi equations to analytically calculate cumulative time-of-flight delays (Δt) for gamma-ray bursts propagating under Quintessence-based FLRW metrics.	
GRB Cosmography & Relativistic Foundations Osservatorio Astronomico di Brera (OAB-INAF)	Merate, Italy June – July 2025
- Constrained global cosmological parameters (Ω_m, Ω_Λ) via χ^2 minimisation profiles using the empirical $E_{\text{peak}} - E_{\text{iso}}$ relation of Gamma-Ray Bursts. - Studied relativistic jet kinematics, special relativity corrections, and detection synergies for Einstein Telescope mock populations under the guidance of Dr. G. Ghisellini, Dr. G. Ghirlanda, and Dr. O. S. Salafia.	
PI GAMMA Project Founder & Project Lead – CAPIBARA Collaboration	International 2024 – Present
Founded an international physics collaboration for science-driven students; leading the GAMMA Team to evaluate scientific parameters and instrumentation requirements for a transient-observing CubeSat.	
Type Ia Supernovae: Mapping Expansion History Institute of Space Sciences (ICE-CSIC-IEEC)	Barcelona, Spain June – July 2024
Modeled non-linear cosmic expansion history, solving the Friedmann acceleration equations to parameterize cosmic density metrics via light-curve parameterisation using the sncosmo suite. <i>Supervisor:</i> Dr. L. Galbany.	
Cepheid Variable Stars Research Research Fellow – Youth & Science Program	Budapest, Hungary Aug – Dec 2023
Reconstructed classical Hubble-Lemaître expansion constraints ($H_0 = 70.26 \pm 7.1 \text{ km s}^{-1} \text{ Mpc}^{-1}$) using observational data from the Konkoly Observatory. Selected for a selective 3-year funded research track.	

AWARDS & HONORS

Youth & Science Fellowship · <i>Fundació Catalunya La Pedrera</i>	2023 – 2026
Fully funded 3-year international research fellowship supporting consecutive competitive summer research placements.	
Abiturpreis (Physics Distinction) · <i>German Physical Society (DPG)</i>	2025
National prize awarded for outstanding academic excellence and foundational focus in physics during high school.	
International Astronomy and Astrophysics Competition (IAAC)	2023 – 2025
Awarded the Silver Honour and the National Award for Spain (2025), ranking in the top 7% globally.	
Jugend Forscht (Youth Research Competition)	2024 – 2025
Awarded First Prize Iberia across regional/national tiers; Special Award in Scientific Photography (NRW, 2025).	

TALKS & SCIENCE OUTREACH

SSEA Symposium (2026): Presented *“The CRD Instrument and the COSMOS Program.”* · **CosmoXarxa** (2025): Invited presentation on high-energy astrophysics metrics at the Barcelona Science Museum · **Workshops** (2023–2024): Designed introductory physics and cosmology seminars for high school tracks at the German School Barcelona.

EMPLOYMENT & TECHNICAL SKILLS

LMU Munich Assistantship: Student Assistant at Diversity Management & Research Funding Unit 2025 – Present

Programming & Tools: Python (*NumPy, SciPy, Pandas, Matplotlib, Astropy*), `sncosmo`, C++, L^AT_EX, Markdown, Git, Linux.